

The first ramjet engine, the GIRD-04, was designed by I.A. Merkulov and tested in April 1933

Green tech 35

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will carry 12 tonnes of cargo into low-earth orbit. Finally, later this year, the US Air Force expects to leave everyone in the dust with Boeing's new X-51 scramjet. All three projects use slightly different techniques to get the most out of their engines, and all three of them must overcome the same hurdle.

The problems

By now, you might have guessed why aircraft manufacturers haven't shown the ramjet or scramjet much regard. For the engines to work, they need to move. Which means that something needs to get them up to supersonic speed before they start operating. More importantly, scramjets aren't likely to take their aircraft to Mach 12 – which is the speed that they need to enter the Earth's orbit – so something also needs to give them that extra push into space.

Here's where the researchers differ the most. The X-43A was dropped into the skies by a B52 bomber, and then used booster rockets to get up to about Mach 4, after which the scramjets kicked into action. All very well for a research aircraft, but we need a plane that can go from runway to stratosphere in a single stage. After the X-43's successful run, NASA's engineers made it clear that they didn't believe that was possible. Their plan is to build an aircraft that'll get to orbit in two stages – most likely a booster rocket for the initial thrust, with the scramjets taking over later. The X-51, too, will fit into this plan.

Reaction Engines, however, has bigger plans for the Skylon. To make the scramjet function at lower speeds, it will use a heat exchanger to cool the incoming air to as low as -100° C. When the fuel (hydrogen, in this case) burns in this cold air, it'll expand rapidly, giving the engines enough thrust to get up in the air and start getting to Mach 5. Finally, at a height of 26 km, it'll switch to good old rocket power for a final push into space.

What Now?

Right now, the future of the scramjet is anyone's guess – it all depends on how researchers plan on tackling the get-it-to-Mach-5 problem. If the Skylon succeeds, it'll score a big point for those

who support the scramjet. If not, however, we come back to the jet-versus-rocket battle that's being fought right now. Meanwhile, we can take a step down to earth.

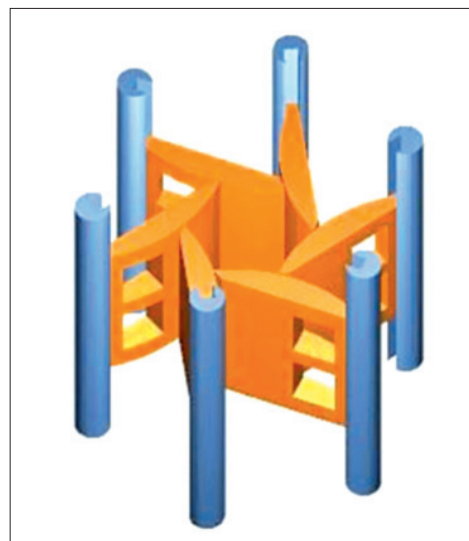
Driving around

For all their fuel-efficiency, the cars we drive are still based on the internal combustion engine (ICE). This is likely to go on for a while, because despite the push for more electric cars, petrol will take you farther, and takes only a few minutes to refill. And, of course, international oil politics will prevent the world from switching to electric cars overnight.

For now, researchers must work on engines that are modifications of the original ICE, or like Tendix, come up with a brand new design for combustion engines.

The IRIS

The biggest problem with the traditional ICE is that it's inefficient – only 25 per cent of the energy that goes into it drives the wheels. The rest goes into overcoming the friction between all the moving parts – the pistons and cylinders, most notably – and gets wasted as heat. This made inventor Timber Dick a sad panda, so last year, he set about designing a more efficient engine, and called it the IRIS – the Internally Radiating Impulse Structure. Conveniently enough, the engine even works like a camera's iris – instead of pistons that go up and down, it features six chordons – door-like structures – arranged in a hexagon. As they rotate, they transfer power to the engine's gears.



A 3D model of the IRIS engine. The inventor's family hopes to create a working prototype soon

If the company manages to get the IRIS off the ground, it'll likely have an efficiency of 70 per cent, which means that you'll be able to get twice as far – if not farther – as you do right now.

The long wait

Don't expect to be flying in a Skylon or X-51 any time soon. NASA's plan for the two-stage-to-orbit aircraft is a 20-year plan, so you're going to be considerably aged on your first flight. The Skylon is still a concept, so we still need to see whether it materialises at all.

As for driving around in a car with an IRIS engine, it all depends on whether the IRIS can make it to the market before...

well... something else does. In any case, the IRIS can't be a long-term solution, because eventually, something needs to replace petrol engines.

Meanwhile, get used to the whine of jet engines, and enjoy a hybrid or two while you're on the roads. ■

readersletters@thinkdigit.com

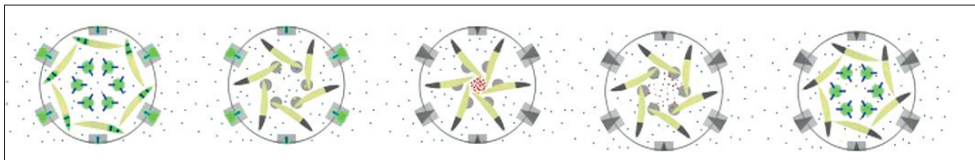
The Case For The Rocket

While everyone's pitching scramjets as the engines of the future, there are several who argue that we shouldn't waste our time on them. The engines of the future have been here all along – rockets. For every one of the scramjet's advantages, the rocket has an answer.

The scramjet promises to reduce the weight that the aircraft carries into orbit – no more booster rockets. However, all the fuel in the booster rockets burns up on the way up, and the boosters themselves are tossed off. In fact, after the boosters go, the aircraft that remains can even be lighter than its scramjet equivalent – NASA found that out when they were studying designs for a single-stage rocket.

In fact, the rocket doesn't seem to have anything against it. Not during a flight, anyway.

The only disadvantage of rockets seems to be the whole production behind each launch – building and filling the boosters, attaching them to the aircraft, and so on. If someone figures out a way to get all that done quicker and cheaper, the scramjet will go back to the history books soon after.



From left to right: In the first phase of the cycle, all the chordons are wide open, and air enters the engine. In the second, the inlets close, and the chordons close inward, compressing the air in the middle. Fuel is then injected into this space, and burned. The hot air now expands, pushing the chordons outward – this is the cycle that does the work. Finally, the exhaust valves open, and let the exhaust out